

MOHAMMAD AKIF AKHTAR

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AI/ML Engineer | Machine Learning | Deep Learning | Computer Vision | NLP | Model Deployment

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Generative AI, NLP, Computer Vision, and RAG systems. Skilled in Python, TensorFlow/Keras, PyTorch, CNN, YOLO, Transformers, embeddings, vector databases, LangChain, and Ollama. Experienced in building RAG pipelines, semantic search, document-based question answering, and LLM-powered applications, along with ML/DL solutions for image classification and prediction. Proficient in deploying AI applications using FastAPI, Flask, Gunicorn, and Render, with experience across the complete pipeline from data preprocessing and model training to evaluation, RAG integration, and production deployment.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Data Structures & Algorithms (DSA)

Machine Learning: Scikit-learn, Random Forest, Logistic Regression, SVM, Decision Tree, Classification, Clustering (K-Means), Feature Engineering, Model Evaluation, Cross-Validation

Deep Learning: TensorFlow, Keras, PyTorch, CNN, YOLO11 (Ultralytics), Neural Networks, Data Augmentation, RNN, LSTM, GRU, Transformers

Generative AI & LLMs: Generative AI, LLMs, RAG, LangChain, Ollama, Hugging Face Transformers, Prompt Engineering, Embeddings, Sentence Transformers, Vector Databases, Semantic Search, Document Q&A

Computer Vision: Image Classification, Image Preprocessing, OpenCV, Pillow

NLP: TF-IDF, Text Classification, Text Preprocessing, Transformers, Embeddings, Semantic Search

Analytics & Data: Pandas, NumPy, Matplotlib, Seaborn, Excel, Power BI, RFM Analysis, Market Basket Analysis (Apriori), Association Rules

APIs & Deployment: FastAPI, Flask, Gunicorn, Render, Model Serialization (Pickle/Joblib)

Tools: Git, GitHub, Jupyter Notebook, VS Code, Google Colab

PROJECTS

Diabetic Retinopathy Detection | CNN, YOLO11, Flask, OpenCV, ReportLab

- Built a YOLO11-based computer vision system to classify diabetic retinopathy severity across 5 clinical stages, achieving 84.97% Top-1 accuracy and 100% Top-5 accuracy on 366 evaluation images.
- Developed a Flask-based inference pipeline with OpenCV/Pillow preprocessing, Top-3 confidence-based predictions, and automated PDF report generation using ReportLab.

Loan Approval Prediction — Machine Learning | Scikit-learn, Random Forest, Flask, Render

- Built a Random Forest classifier to predict loan approval, achieving 96.8% accuracy, 0.96 F1-score, and 0.97 ROC-AUC, outperforming Logistic Regression and Decision Tree baselines.
- Serialized the model with Joblib and deployed a Flask web app live on Render for real-time predictions.

Customer Segmentation — Unsupervised Learning | Scikit-learn, K-Means, Apriori

- Performed RFM (Recency, Frequency, Monetary) analysis on e-commerce transaction data and applied K-Means clustering to segment customers into 4 actionable groups.
- Applied Market Basket Analysis using the Apriori algorithm to mine frequent itemsets and generate association rules (support, confidence, lift).

Leaf Disease Prediction — Deep Learning | TensorFlow, Keras, CNN

- Trained a custom CNN from scratch using TensorFlow/Keras to classify 15 leaf disease classes across Bell Pepper, Potato, and Tomato crops, achieving 96.06% validation accuracy and 95.04% macro F1-score.
- Applied data augmentation (flip, rotation, zoom, contrast) and image preprocessing, using EarlyStopping, ModelCheckpoint, and ReduceLROnPlateau to optimize training.

Email Spam Classifier — NLP | TF-IDF, SVM, Scikit-learn

- Built an NLP pipeline (text cleaning, stopword removal, stemming/lemmatization, TF-IDF vectorization) to classify emails as spam or legitimate, evaluating Logistic Regression, Naive Bayes, SVM, and Random Forest.
- Selected SVM as the top performer (99.21% accuracy) and built a Flask web interface returning spam probability and risk level in real time.

EXPERIENCE

Samatrix Consulting Private Limited — ML/Data Analysis Intern 01 Jul 2026 – 28 Jul 2026

- Performed Python-based financial data preprocessing, statistical analysis, and visualization for quantitative portfolio risk assessment.
- Calculated 1-Day 95% Value at Risk (VaR) to evaluate potential portfolio losses and support risk-based decision-making.
- Applied data analysis and machine learning techniques to financial datasets to identify patterns and generate actionable insights.

ACHIEVEMENTS

- Led a team to win the Best Visionary Team Award at an intra-university hackathon for DisasterSense, an AI-based disaster response/prediction project.

EDUCATION

B.Tech — Computer Science & Engineering (AI & ML), Quantum University 2024 – 2028

CGPA: 8.25/10